

Earners AI and ML by Aditya Jain sir - Lecture 15

Summary Notes

Recap - Tuple

```
a = (1,4,6,3,7)
type(a)
```

tuple

```
len(a)
```

5

```
a[2]
```

6

```
a[2] = 10
```

```
-----
TypeError                                Traceback (most recent call last)
/tmp/ipykernel_430/3576902070.py in <cell line: 0>()
----> 1 a[2] = 10

TypeError: 'tuple' object does not support item assignment
```

Next steps: [Explain error](#)

```
n = len(a)
for i in range(n):
    print(a[i])
```

1
4
6
3
7

```
for elem in a:
    print(elem)
```

1
4
6
3
7

```
a = ("Raj" , 24)

name, age = a
print(name)
print(age)
```

Raj
24

```
a = (2,)
len(a)
```

1

```
type(a)
```

tuple

HW

```
values = (5, 10, 15, 20)

total = 0

for value in values:
    if value % 5 == 0:
        total += value // 5

print(total)
```

10

```
colors = ("Red", "Blue", "Green")

for color in colors:
    print(color[0], end=" ")

print()
```

R B G

```
numbers = (1, 2, 3)

a, b, c = numbers

print(a + b * c)
```

7

```
marks = (80, 90, 95)

bonus = ()

for mark in marks:
    bonus += (mark + 5,)

print(bonus)
```

(85, 95, 100)

```
fruits = ("Apple", "Banana", "Mango")

for fruit in fruits:
    if "a" in fruit.lower():
        print(fruit)
```

Apple
Banana
Mango

"Aditya".lower()

'aditya'

"Aditya".upper()

'ADITYA'

```
fruits = ("Apple", "Banana", "Mango")

for fruit in fruits:
    if "a" in fruit:
        print(fruit)
```

Banana
Mango

```
prices = (100, 200, 150)

discount = ()
```

```
for p in prices:
    discount += (p - 20,)

print(sum(discount))
```

390

```
tuple1 = (1, 2)

tuple2 = (3, 4)

tuple3 = tuple1 + tuple2

print(tuple3 * 2)
```

(1, 2, 3, 4, 1, 2, 3, 4)

(1,2)*3

(1, 2, 1, 2, 1, 2)

"a"*5

'aaaaa'

[1,2]*2

[1, 2, 1, 2]

```
data = (2, 4, 6, 8)

count = 0

for num in data:
    if num % 4 == 0:
        count += 1

print(count)
```

2

▼ DPP - Module 1 Challenge

```
numbers = [10, 15, 20, 25]

result = ()

for num in numbers:
    if num % 10 == 0:
        result += (num // 2,)

print(result)
print(sum(result))
```

(5, 10)
15

```
text = "Programming"

count = 0

for ch in text:
    if ch.lower() in "aeiou":
        count += 1

print(count)
```

3

#Given a string, write a code to convert it to lowercase. U cannot use lower()

```
s = "Raj"
# op: "raj"

new_s = ""
for ch in s:
    if ch <= 'Z' and ch >= 'A':
        new_s += chr(ord(ch)+32)
    else:
        new_s += ch

print(new_s)
```

raj

```
#Given a string, write a code to convert it to uppercase. U cannot use upper()
s = "Raj"
# op: "raj"

new_s = ""
for ch in s:
    if ch <= 'z' and ch >= 'a':
        new_s += chr(ord(ch)-32)
    else:
        new_s += ch

print(new_s)
```

RAJ

```
print(ord("A") , ord("a"))

ord("a") - ord("A")
```

65 97
32

```
student = {
    "Math":90,
    "Science":80,
    "English":95
}

total = 0

for subject in student:
    if student[subject] > 85:
        total += student[subject]

print(total)
```

185

```
marks = [80, 90, 75]

def average(data):
    return sum(data)/len(data)

print(average(marks))
```

81.66666666666667

```
numbers = [3,5,7,8,10]

count = 0

for n in numbers:
    if n % 2 == 0:
        continue

    count += 1

print(count)
```

3

```
names = ["Amit", "Rahul", "Riya", "Neha"]

largest = ""

for name in names:
    if len(name) > len(largest):
        largest = name

print(largest)
```

Rahul

```
numbers = [5,10,15,20]

result = []

for n in numbers:
    if n>10:
        result.append(n)

print(result)
print(len(result))
```

[15, 20]
2

```
text = "banana"

count = {}

for ch in text:
    if ch not in count:
        count[ch]=1
    else:
        count[ch] += 1

print(count)
```

```
{'b': 1}
{'b': 1, 'a': 1}
{'b': 1, 'a': 1, 'n': 1}
{'b': 1, 'a': 2, 'n': 1}
{'b': 1, 'a': 2, 'n': 2}
{'b': 1, 'a': 3, 'n': 2}
```

```
print(count)
```

```
{'b': 1, 'a': 3, 'n': 2}
```

```
numbers = (5,10,15)

result = 0

for n in numbers:
    result += n

print(result//len(numbers))
```

10

```
student = {
    "Math":80,
    "Science":90
}

student["English"]=95

total = 0

for sub in student:
    total += student[sub]

print(total)
```

265

```
numbers = [2,4,6]

product = 1

for n in numbers:
    product *= n

print(product)
```

48

```
def fun(word):
    return word[::-1]

print(fun("Python"))
```

nohtyP

```
s = "Python"

print(s[4:1:-1])

oht
```

```
numbers = [1,2,3,4]

result = []

for n in numbers:

    if n%2==0:
        result.append(n*n)

print(result)
```

[4, 16]

```
numbers = [1,2,3,4]

result = []

for n in numbers:
    result.append(n*n)
    print(result)
```

```
[1]
[1, 4]
[1, 4, 9]
[1, 4, 9, 16]
```

```
marks = {
    "A":50,
    "B":60,
    "C":70
}

highest = 0
```

```
for key in marks:

    if marks[key]>highest:
        highest=marks[key]

print(highest)
```

70

```
text = "Data Science"

count = 0

for ch in text:

    if ch==" ":
        continue

    count+=1

print(count)
```

11

```
len(text)
```

12

```
students = [
    "Rahul",
    "Amit",
    "Neha"
]

for i in range(len(students)):

    print(i+1, students[i])
```

```
1 Rahul
2 Amit
3 Neha
```

```
numbers = [2, 5, 8, 11, 14]

result = []

for num in numbers:
    if num % 2 == 0:
        result.append(num * 2)
    else:
        result.append(num + 1)
    print(result)

print(sum(result))
```

```
[4]
[4, 6]
[4, 6, 16]
[4, 6, 16, 12]
[4, 6, 16, 12, 28]
66
```

```
text = "Python Programming"

count = 0

for ch in text:
    # if ch == " ":
    #     continue
    if ch.lower() in "aeiou":
        count += 1

print(count)
```

4

```
marks = [85, 90, 75, 95, 80]

highest = marks[0]

for mark in marks:
    if mark > highest:
        highest = mark

marks.remove(highest)

print(max(marks))

90
```

▼ DPP

```
def fun(values):

    total = 0

    for v in values:
        if v % 2 == 0:
            total += v

    return total

numbers = [5, 8, 12, 7, 10]

print(fun(numbers))
```

```
student = {
    "Math": 80,
    "Science": 90,
    "English": 85
}

count = 0

for subject in student:
    if student[subject] >= 85:
        count += 1

print(count)
```

```
names = ["Rahul", "Amit", "Neha", "Riya"]

answer = ""

for name in names:
    if len(name) == 4:
        answer += name[-1]

print(answer)
```

```
numbers = (10, 15, 20, 25)

result = ()

for n in numbers:
    if n % 5 == 0:
        result += (n // 5,)

print(result)
```

```
values = {2, 4, 6, 8}

total = 0

for value in values:
```



```
    total += value

print(total // len(values))
```

```
numbers = [3, 6, 9]

new = []

for n in numbers:
    new.append(n * n)

print(max(new))
```

```
student = ("Rahul", 22, 95)

name, age, marks = student

if marks > 90:
    grade = "A"
else:
    grade = "B"

print(name, grade)
```

```
fruits = ["Apple", "Banana", "Mango"]

result = []

for fruit in fruits:
    if len(fruit) > 5:
        result.append(fruit.upper())

print(result)
```

```
data = {
    "A": 5,
    "B": 10,
    "C": 15
}

total = 0

for key in data:
    total += data[key]

print(total / len(data))
```

```
numbers = [2, 4, 6, 8]

i = 0

total = 0

while i < len(numbers):
    total += numbers[i]
    i += 2

print(total)
```

```
text = "Mississippi"

count = {}

for ch in text:
    if ch not in count:
        count[ch] = 1
    else:
        count[ch] += 1

print(count["s"])
```

```
words = ["cat", "elephant", "dog", "tiger"]
```

```
longest = words[0]

for word in words:
    if len(word) > len(longest):
        longest = word

print(longest)
```

```
students = {
    "Rahul": 90,
    "Amit": 75,
    "Neha": 95
}

highest_name = ""

highest_marks = 0

for name in students:
    if students[name] > highest_marks:
        highest_marks = students[name]
        highest_name = name

print(highest_name)
```